Mohammed VI Polytechnic University

College of Agriculture and Environmental Sciences

International Water Research Institute (IWRI)

organizes a workshop:

AI-Based Simulation-Optimization for Solving Groundwater Flow Problems

Dr. Andy Louwyck
Dr. Stefaan Haspeslagh

April 7 and 8, 2026

Seminar Room : Asmara

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AI-Based Simulation-Optimization for Solving Groundwater Flow Problems

This two-day workshop provides an integrated introduction to methods for modeling and optimizing groundwater flow systems.

Tuesday, April 7

09:00–18:00: Simulation of Groundwater Flow to Wells – Dr. Andy Louwyck

The first day focuses on fundamental and applied aspects of groundwater flow toward pumping wells. Topics include fundamental principles of groundwater flow theory, analytical and numerical formulations, interpretation of aquifer tests, and the use of Python for implementing groundwater flow models.

Wednesday, April 8

09:00–18:00: Optimization of Groundwater Flow Problems – Dr. Stefaan Haspeslagh

The second day introduces a range of optimization techniques relevant to groundwater management. We examine linear programming, metaheuristic approaches such as simulated annealing and genetic algorithms, and local search strategies. Emphasis is placed on how these methods can be coupled with simulation models for groundwater flow to support decision-making in groundwater abstraction and resource management.

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About the Instructors

Andy Louwyck holds a PhD in Earth Sciences from Ghent University and has over twenty years of experience in research, policy studies, consulting, and higher education. He is AI Lead at the Flanders Environment Agency and lecturer in Applied Computer Science at VIVES University of Applied Sciences. His expertise includes hydrogeology, groundwater modeling, scientific computing, environmental data science, machine learning, and artificial intelligence. He has published in leading international journals and works across scientific, technical, and applied domains.

Stefaan Haspeslagh holds a PhD in Computer Science from KU Leuven and specializes in combinatorial optimization, spanning metaheuristic, exact, and hybrid methods. Since 2012, he has been a lecturer in Applied Computer Science at VIVES University of Applied Sciences, where he has led applied AI research, developed AI curricula, and coordinated translational projects across healthcare, industry, education, and logistics. As Innovation Manager AI, he strengthens collaborations with KU Leuven and industry and develops joint research and valorization initiatives.

This two-day workshop offers an intensive, hands-on learning experience. Each day runs from 9:00 to 18:00, with dedicated breaks from 10:30–10:45, 12:30–15:00, and 16:30–16:45, for a total of six hours of interactive teaching. Participants are invited to bring their own laptops to fully engage with the practical exercises. All Python examples are provided in Jupyter notebooks and can be executed easily and free of installation through Google Colab.

The workshop is organized by the International Water Research Institute (IWRI):

Dr. Athmane Khettouch
Dr. Imane El Bouazzaoui
Prof. Dr. Yassine Ait Brahim

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